

Flight System Requirements

Oskar Meszaros (z5636164)

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1 Introduction

This document outlines the requirements of all flight systems. The scope of this document includes all electrical and software requirements pertaining to relevant systems within the Flight Systems subteam.

1.1 Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" are to be interpreted as in [RFC 2119](#). Note that these definitions are relative to the board or system that it is concerned with. For example, the Pfozter-Regener detector must include a silicon photomultiplier. This means the Pfozter-Regener subsystem requires a silicon photomultiplier to function correctly, however this isn't a strict requirement of the project overall as the Pfozter-Regener subsystem is a should have requirement.

This can be summarised with

- **MUST, REQUIRED, SHALL** mean that the definition is an absolute requirement
- **MUST NOT, SHALL NOT** mean that the definition is an absolute prohibition.
- **SHOULD, RECOMMENDED** mean that there may exist valid reasons in particular circumstances to ignore a particular item or definition, but the full implications must be understood.
- **SHOULD NOT, NOT RECOMMENDED** mean that there may exist valid reasons in particular circumstances when a particular item or definition is acceptable, but the full implications must be understood.
- **MAY, OPTIONAL** mean that an item is truly optional.

Every requirement is assigned a level, indicating the degree to which any particular definition is required for the success of the Payload.

1. A level 1 requirement indicates a "hard" requirement. These define the bare minimum in order to have functioning payload, as well as defining any limits associated with the payload.
2. A level 2 requirement indicates an "good to have" requirement. These define systems or definitions that are good to have on a functional payload.
3. A level 3 requirement indicates an "ideal" requirement. These are systems or definitions that would be present on an ideal payload.

Additionally, all requirements have an associated ID. This takes the form of **RL.XX/YY-ZZ**, where L is the level of the requirement, **XX** denote the major number, **YY** is the minor letter/number, and **ZZ** is the particular index. Note that all numbers are 0-padded, and that any field may be omitted (bar the major number). For example, the requirement dictating the required voltage for the experimental payload beds is **R2.07/B**, and the particulars of the requirement can be referred to as **R2.07/B-i** and **R2.07/B-ii**, respectively.

2 High-Level Design

2.1 Overview

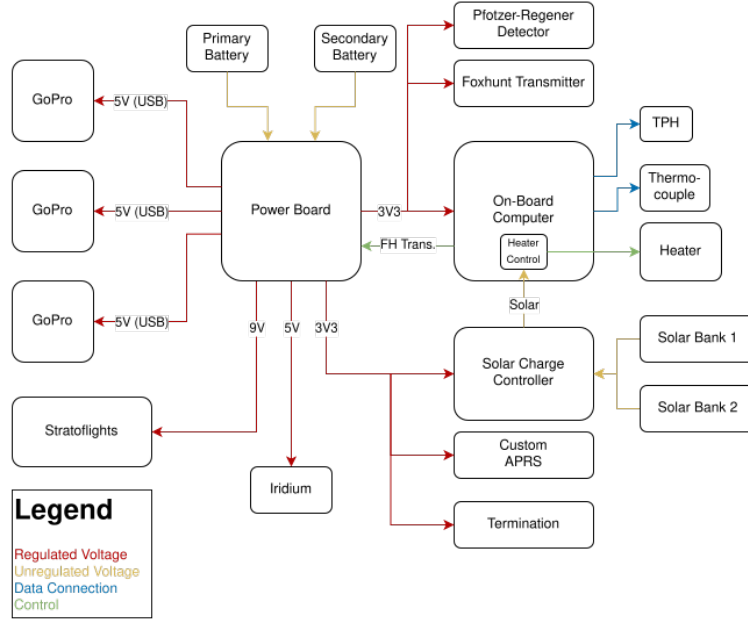


Figure 1: The high-level block diagram for the V2 payload.

The flight systems on the High-Altitude Balloon (HAB) is defined as any electrical system that does not require communication with any satellite or receiver that isn't GPS/GNSS. On the V2 payload, this includes the On-Board Computer (OBC), Power Management Board (PMB), and On-Board Flight Cameras (OBFC). Also included are the two experimental subsystems, the Pfofzer-Regener Detector (PRD) and Solar Monitoring Board (SMB).

2.2 Requirements

The below dictates the hard (or level 1) requirements of the Payload.

1. The total cost of Payload must not exceed \$2000, excluding existing components, materials, and tools
2. The total weight of the Payload must not exceed 2.5kg
3. The Payload must be entirely contained within the as yet to be defined volume of the Payload. This excludes
 - (a) Any external sensors, such as TPH
 - (b) Any external devices that require outside access, such as the OBFC
4. All devices of the Payload must be recovered and should be in a state to be reused for future launches.
 - (a) The electronics must be capable of withstanding impact at a peak descent velocity of 5m/s
5. The electronics of the Payload must be active once given an appropriate power source
6. The Payload must be easily locatable and visible once it has landed
 - (a) The Payload must be brightly coloured

- (b) The Payload must have a radio beacon that transmits a constant radio signal post-landing
 - i. The beacon must continuously transmit for a minimum of 5 hours post-landing
 - ii. The beacon must have a minimum range of 200m in all directions
- 7. The Payload must have a dedicated experimental payload bed
 - (a) The Payload bed must be capable of mounting an experimental payload with dimensions no greater than $100 \times 100 \times 100\text{mm}$
 - (b) The Payload must supply a regulated voltage rail to the experimental payload bed
 - i. The specific voltage/capacity of each experimental bed power rail is dictated by the experiment itself
 - ii. A particular experiment may not require an external power source, and in such a circumstance one will not have to provided by the Payload
- 8. The Payload must have temperature, pressure, and humidity sensors both within and immediately surrounding the Payload, operational for the duration of the flight
 - (a) The Payload must have at least 1 primary and 1 secondary TPH, both within and immediately surrounding external to the Payload
 - (b) All TPH sensors must provide measurements at a frequency of no greater than every 5 seconds
 - (c) All temperature sensors must provide measurements with an accuracy of no greater than $\pm 1^\circ\text{C}$
 - (d) All pressure sensors must provide measurements with an accuracy of no greater than $\pm 2\text{hPa}$
 - (e) All humidity sensors must provide measurements with an accuracy of no greater than $\pm 5\%\text{RH}$
 - (f) All temperature, pressure, and humidity sensors must be capable of reliably operating with an external Payload ambient temperature of -70°C
- 9. The Payload must be capable of measuring the altitude of the Payload for the duration of the flight
 - (a) The Payload must have at least 1 primary and 1 secondary altitude measurement unit within the Payload
 - (b) All altitude measurement units must provide measurements at a frequency of no greater than every 5 seconds
 - (c) All altitude measurements must provide measurements with an accuracy of no greater than $\pm 20\text{m}$
 - (d) All altitude measurement units must be capable of reliably operating with an external payload ambient temperature of -70°C
- 10. The Payload must have 9-axis (accelerometer, gyroscope, magnetometer) inertial measurements units (IMUs) within the Payload, operational for the duration of the flight
 - (a) The Payload must have at least 1 primary and 1 secondary 9-axis IMU within the Payload
 - (b) All IMUs must provide measurements at a frequency of no greater than every 10ms
 - (c) All IMU accelerometers must provide measurements with a range of at least $\pm 16\text{g}$
 - (d) All IMU gyroscopes must provide measurements with a range of at least $\pm 2000\text{dps}$
 - (e) All IMU magnetometers must provide measurements with a range of at least $\pm 1300\mu\text{T}$
 - (f) All IMUs must be capable of reliably operating with an external Payload temperature of -70°C
- 11. The Payload must have reliable and continuous two-way communication to a ground-based team for the duration of the flight
- 12. The Payload have at least 3 cameras operational for the duration of the flight
 - (a) All cameras must record at a resolution of at least 1080p

- (b) All cameras must record at a frame rate of at least 30 frames per second
 - (c) All cameras must have sufficient storage capacity to store all recorded footage for the duration of the flight
 - (d) All cameras must be external-facing and have a minimum horizontal field of view of 120° for the duration of the flight
 - (e) All cameras must be capable of reliably operating with an external Payload ambient temperature of -70°
 - (f) Collectively, all cameras should cover the following angles
 - i. Vertically to capture the balloon
 - ii. Vertically to capture the earth
 - iii. Horizontally to capture the horizon
13. The Payload must store all primary and secondary flight environment and flight dynamics data from the duration of the flight

3 On-Board Computer

3.1 Overview

The On-Board Computer (OBC) is the main controller for the payload. It is directly responsible for the control of the battery heater(s), control of the foxhunt transmitter, and logging temperature (TPH) and acceleration, rotation, and magnetism.

3.2 Requirements

4 Power Management Board

4.1 Overview

4.2 Requirements

5 On-Board Flight Cameras

5.1 Overview

5.2 Requirements

6 Pftotzer-Regener Detector

6.1 Overview

6.2 Requirements

7 Solar Monitoring Board

7.1 Overview

7.2 Requirements

A Glossary of Terms

B Specifications of Parts